

Victor Maull Miquel

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Google Scholar (89 citations, h-index 5)
Last updated: July 2026

PROFILE

I am a theorist and experimentalist working at the interface of complex systems science, theoretical and field ecology and synthetic biology. My research focuses on the emergence of ecological organization across scales, from microbial communities to the biosphere, and on the role of feedbacks, critical transitions and ecosystem engineering in shaping ecological dynamics. I combine mathematical modelling, synthetic ecosystems and field studies to understand and design interventions for biodiversity conservation, ecosystem restoration and planetary stewardship.

POSITIONS

2026–	Universitat Pompeu Fabra, UPF , Barcelona, Spain. Post Doctoral Researcher. Supervisor: Ricard Solé.
2023–2026	Museum of Natural Sciences of Barcelona , Barcelona, Spain. Project Manager, <i>MicrocosmosBCN</i> . Funded by the Barcelona City Council and the “la Caixa” Foundation.
2019-2026	Universitat Pompeu Fabra, UPF , Barcelona, Spain. Pre Doctoral Researcher. Supervisor: Ricard Solé.
2016-2017	Institute of Marine Sciences, ICM , Barcelona, Spain. Undergraduate Researcher. Supervisor: Dolors Vaqué.

EDUCATION

2026	PhD in Complex Systems , Universitat Pompeu Fabra (UPF). Thesis: “Synthetic Ecosystems: Design Directions for Restoration and Conservation”. <i>Excellent Cum Laude</i> . Advisor: Ricard Solé.
2022	Complex Systems Summer School (CSSS) , Santa Fe Institute (SFI), Santa Fe, NM, USA. Selected participant.
2019	MSc in Bioengineering , Chemical Institute of Sarrià (IQS).
2018	International Genetically Engineered Machine (iGEM) Competition , Boston, MA, USA. Team Leader of the Bio IQS Barcelona team. Bronze Medal.
2017	BSc in Biological Systems Engineering , Polytechnic University of Catalonia.
2013	BSc in Industrial Engineering (unfinished), Polytechnic University of Catalonia.

PUBLICATIONS ^{*}*Equal contribution*

[†]Corresponding author

- V. Maull**[†], Y. Shpilkin, V. de Lorenzo, R. Solé, “Emergent bioengineering”, *Current Opinion in Biotechnology*, 100, 103520 (2026).
- R. Solé, **V. Maull**, D. R. Amor, J. P. Mauri, N. C. Pueyo, “Synthetic ecosystems: from the test tube to the biosphere”, *ACS Synthetic Biology*, 13(12), 3812–3826 (2024).
- V. Maull**, R. Solé, “Biodiversity as a firewall to engineered microbiomes for restoration and conservation”, *Royal Society Open Science*, 11(6), 231526 (2024).
- V. Maull**, J. P. Mauri, N. C. Pueyo, R. Solé, “A synthetic microbial Daisyworld: planetary regulation in the test tube”, *Journal of the Royal Society Interface*, 21(211), 20230585 (2024).
- V. Maull**, J. P. Mauri, N. C. Pueyo, R. Solé, “Planetary regulation in the test tube: a synthetic Daisyworld”, *Artificial Life Conference Proceedings*, 36(1), 67 (2024).
- V. Maull**, R. Solé, “Network-level containment of single-species bioengineering”, *Philosophical Transactions of the Royal Society B*, 377 (2022).
- B. Vidiella, A. Guillamon, J. Sardanyés, **V. Maull**, J. Pla, N. Conde, R. Solé, “Engineering self-organized criticality in living cells”, *Nature Communications*, 12(1), 4415 (2021).
- R. Solé, N. Conde-Pueyo, A. Guillamon, **V. Maull**, J. Pla, J. Sardanyés et al., “Synthetic criticality in cellular brains”, *Journal of Physics: Complexity*, 2(4), 041001 (2021).

PREPRINTS

- J. Pla-Mauri^{*}, **V. Maull**^{*}, A. Tabi, Y. Shpilkin, V. de Lorenzo, R. Solé, “Ecological constraints to mirror life”, *bioRxiv*, 2026.05.07.723461 (2026).
- V. Maull**[†], G. Aguadé-Gorgorió, V. de Lorenzo, R. Solé, “Synthetic Horizontal Gene Transfer for Ecosystem Restoration”, *bioRxiv*, 2025.09.23.678013 (2025).

RESEARCH VISITS

- 2026 **Ascension Island**, South Atlantic Ocean (UK Overseas Territory). Visiting Researcher hosted by the Ascension Heritage Society and the Ascension Island Government Conservation and Fisheries Directorate (AIGCFD) (March–April 2026).
- 2025 **King Abdullah University of Science and Technology (KAUST)**, Thuwal, Saudi Arabia. Visiting Researcher in the laboratory of Fernando T. Maestre (2 weeks).
- 2025 **Institute of Evolutionary Sciences of Montpellier (ISEM)**, Montpellier, France. Visiting Researcher in the laboratory of Sonia Kéfi (1 month).
- 2019–2024 **Santa Fe Institute (SFI)**, Santa Fe, NM, USA. Visiting Researcher during six research stays (1–2 months each), including participation in the Emergent Engineering - Foundations Working Group in 2024.

AWARDS FUNDING	&	Competitive funding	
		2026	ReTRaCaT Grant for the Promotion of Research and Scientific Dissemination , Universitat Pompeu Fabra and AGAUR (€1,150). Awarded competitive funding for the writing and dissemination of the popular science book project <i>Garden on a Cinder: Ascension Island and the Origins of Ecosystem Engineering</i> .
		2025	Erasmus+ Traineeship Grant , European Commission. Awarded mobility funding for a research stay in the laboratory of Sonia Kéfi, University of Montpellier, France.
		2023	Research and Innovation Grants in Collaboration with the “la Caixa” Foundation , Barcelona City Council and the “la Caixa” Foundation (€150,000). Awarded funding for the project <i>MicrocosmosBCN</i> . Co-author of the proposal and project lead.
		2023	Planetary Wellbeing Initiative Pilot Project , Universitat Pompeu Fabra (€20,000). Awarded funding for the project <i>Terraforming Earth Ecosystems in the Anthropocene Context</i> . Co-author of the proposal and project lead.
INVITED TALKS	&	2020	FI-SDUR Predoctoral Fellowship , Agency for Management of University and Research Grants (AGAUR), Government of Catalonia. Awarded four-year competitive funding for doctoral studies in Complex Systems at Universitat Pompeu Fabra.
		2026	Institute of Natural Resources and Agrobiology of Seville (IRNAS-CSIC) , Seville, Spain. “Ascension Island and the Origins of Ecosystem Engineering”. Invited seminar hosted by Manuel Delgado-Baquerizo at the Biodiversity and Ecosystem Functioning Laboratory (BioFunLab).
		2026	Ascension Heritage Society and Ascension Island Government Conservation and Fisheries Directorate (AIGCFD) , Ascension Island, South Atlantic Ocean (UK Overseas Territory). “Ascension Island and the Origins of Ecosystem Engineering”. Invited lecture hosted by the Government of Ascension Island.
		2025	King Abdullah University of Science and Technology (KAUST) , Thuwal, Saudi Arabia. “Garden on a Cinder: Can Ecosystems be Greened?”. Invited seminar hosted by Fernando T. Maestre.
		2025	BioClub , Tokyo, Japan. “Mapping the Road to Synthetic Ecosystems”. Invited seminar.
		2025	Institut des Sciences de l’Évolution de Montpellier (ISEM) , Montpellier, France. “Synthetic Microbial Daisyworld: Planetary Regulation in the Test Tube”. Invited seminar hosted by Sonia Kéfi.
		2025	GdR Workshop: Incorporating Non-Random Structure into Species-Rich Ecological Models . “Rein Control in Ecology”. Invited workshop contribution hosted by Guim Aguadé-Gorgorió.
		2024	Estación Biológica de Doñana (EBD-CSIC) , Seville, Spain. “Synthetic Microbial Daisyworld”. Invited seminar hosted by Violeta Solanas Calleja and Óscar Godoy del Olmo.
		2024	Barcelona Collaboratorium for Modelling and Predictive Biology , Barcelona, Spain. “Synthetic Microbial Daisyworld”. Invited seminar hosted by the Complex Systems Lab seminar series.

CONFERENCE TALKS	2025	Artificial Life Conference (ALIFE 2025) , Kyoto, Japan. “Mapping the Road to Synthetic Ecosystems”. Contributed oral presentation.
	2024	Artificial Life Conference (ALIFE 2024) , Copenhagen, Denmark. “Planetary Regulation in the Test Tube: A Synthetic Daisyworld”. Contributed oral presentation.
	2024	L24EEDS Workshop: Mathematical Modelling of Microbial Communities: Cooperation, Dynamics and Resistance , Leeds, United Kingdom. “Synthetic Microbial Daisyworld”. Contributed oral presentation.
POPULAR SCIENCE & OUTREACH	2026	The Cybernetic Seed , Museum of Natural Sciences of Barcelona, Barcelona, Spain. Creator of an artistic-scientific installation for the exhibition <i>The Invention of Time</i> .
	2026–	Garden on a Cinder: Ascension Island and the Origins of Ecosystem Engineering . Popular science book project currently in preparation.
	2026	EL PAÍS (forthcoming). Feature article and photo essay documenting field research on Ascension Island and ecosystem restoration.
	2026	Cadena SER , Spain. Interviewee in “A vivir que son dos días” discussing Ascension Island, ecosystem restoration and ecosystem engineering.
	2025	3Cat , Catalonia, Spain. TV interview: “Com estudiar l’emergència climàtica en l’ecosistema peculiar d’un terrat de Barcelona”.
	2024	Visions de la Ciència , Libraries of Barcelona and Barcelona City Council, Barcelona, Spain. “Is it Possible to Redesign the Biosphere?”. Invited public lecture.
SUPERVISION	2025	Yelyzaveta Shpilkina , MSc in Advanced Microbiology, Universitat de Barcelona and Universitat Pompeu Fabra. Supervisor of the Master’s thesis “Microbial Daisyworld: A Synthetic Biology Experiment Toward Testing Gaia Theory”.
METHODS	Complex systems theory, ecological and Earth system modelling, stochastic differential equations and nonlinear dynamical systems, critical transitions, resilience theory and tipping points, bifurcation analysis and alternative stable states, soil–vegetation–atmosphere feedbacks and ecosystem engineering, statistical inference, parameter estimation and uncertainty quantification, geospatial databases, spatial modelling and restoration mapping, numerical simulations and computational modelling, scientific programming (Python, R, Julia), and open and reproducible science practices.	
LANGUAGES	English (fluent), Spanish (native), Catalan (native).	